Name: Prathamesh Sunil Lad
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1. Threat Hunting Methodologies

Overview

Threat hunting is a proactive cybersecurity practice focused on identifying hidden threats that evade traditional security tools. Unlike reactive incident response, which deals with known alerts, threat hunting is hypothesis-driven and aims to detect advanced attacks early.

Core Concepts

Proactive Threat Hunting:

This approach involves creating hypotheses based on attacker behavior and testing them against available data. For example, identifying misuse of valid accounts (MITRE ATT&CK Technique T1078) by analyzing abnormal privilege escalation activities in system logs.

Reactive vs Proactive Approach:

Reactive security responds to alerts after detection, whereas proactive hunting actively searches for suspicious activities before alerts are triggered. This significantly reduces detection time and potential damage.

Threat Hunting Frameworks:

- **SqRR (Search, Query, Retrieve, Respond):** A structured approach to systematically investigate threats
- **TaHiTI (Targeted Hunting integrating Threat Intelligence):** Combines threat intelligence with hunting strategies to improve detection accuracy

Data Sources for Threat Hunting:

Effective hunting requires analyzing multiple data sources, including:

- Endpoint Detection and Response (EDR) logs
- Network traffic data
- System event logs
- Threat intelligence feeds

2. Advanced SOAR Automation

Overview

Security Orchestration, Automation, and Response (SOAR) is used in modern SOC environments to automate repetitive tasks, streamline workflows, and improve incident response efficiency. It reduces manual effort and ensures faster, consistent handling of security incidents.

Core Concepts

SOAR Components:

SOAR consists of three main components:

- **Orchestration:** Integration of multiple security tools and systems to work together in a unified workflow
- **Automation:** Execution of repetitive tasks automatically, such as alert enrichment or ticket creation
- **Response:** Automated or semi-automated actions taken to mitigate threats, such as isolating systems or blocking malicious IPs

Playbook Development:

Playbooks are predefined workflows designed to handle specific types of incidents. For example, in a phishing attack:

- Check email indicators
- Validate IP/domain reputation
- Block malicious sources
- Generate an incident ticket

A practical example includes automating IP blocking for Command and Control (C2) traffic.

Integration with SIEM and EDR:

SOAR platforms are integrated with tools like Wazuh and Elastic Security to enable seamless data flow. Alerts generated in SIEM tools can trigger automated actions through SOAR, reducing response time and improving accuracy.

3. Post-Incident Analysis and Continuous Improvement

Overview

Post-incident analysis is a critical phase in cybersecurity operations that focuses on understanding the root cause of incidents and improving future response strategies. Instead of just resolving an issue, this process ensures that similar incidents are prevented and detection capabilities are strengthened over time.

Core Concepts

Root Cause Analysis (RCA):

Root Cause Analysis is used to identify the underlying reason behind a security incident. Techniques such as the 5 Whys and Fishbone Diagram help in systematically analyzing contributing factors. For example, a phishing attack may reveal weak email filtering mechanisms or lack of user awareness.

Lessons Learned Process:

After resolving an incident, a structured review is conducted to evaluate what worked and what failed. This includes improving detection rules, updating response procedures, and enhancing user training programs.

Metrics and KPIs:

Security performance is measured using key metrics such as:

- **Mean Time to Detect (MTTD):** Time taken to identify a threat
- **Mean Time to Respond (MTTR):** Time taken to mitigate the threat

4. Adversary Emulation Techniques

Overview

Adversary emulation is a proactive cybersecurity practice that involves simulating real-world attacker behavior to evaluate the effectiveness of detection and response mechanisms. It helps organizations understand how attackers operate and identify gaps in their security controls before an actual breach occurs.

Core Concepts

Adversary Emulation:

This technique focuses on replicating attacker tactics, techniques, and procedures (TTPs) based on frameworks like MITRE ATT&CK. Examples include:

- **T1566 (Phishing):** Simulating spear-phishing emails to test user awareness and email filtering systems
- **T1210 (Exploitation of Remote Services):** Simulating attacks on vulnerable services to evaluate system defenses

Emulation Frameworks:

Tools such as MITRE Caldera are used to automate adversary simulations. These frameworks allow mapping of attack techniques to real-world scenarios and help in testing detection rules. For example, a spear-phishing simulation can be executed to evaluate whether security tools can detect and block malicious emails.

Red-Blue Team Collaboration:

Adversary emulation bridges the gap between offensive (Red Team) and defensive (Blue Team) security. Red Teams simulate attacks, while Blue Teams monitor, detect, and respond. This collaboration improves detection capabilities, strengthens defense strategies, and enhances overall SOC readiness.

5. Security Metrics and Executive Reporting

Overview

Security metrics and executive reporting are essential for evaluating SOC performance and communicating outcomes to management. While technical teams focus on detection and response, leadership requires clear, measurable insights to understand risks, performance, and areas for improvement.

Core Concepts

Advanced SOC Metrics:

Security Operations Centers use various metrics to measure effectiveness, including:

- **Dwell Time:** Time between system compromise and detection
- **False Positive Rate:** Percentage of alerts incorrectly identified as threats
- **Incident Resolution Rate:** Number of incidents successfully resolved within a defined time

These metrics help in identifying weaknesses in detection and response mechanisms.

Executive Reporting:

Security findings must be translated into simple, non-technical language for stakeholders. Effective reporting includes:

- Clear summaries of incidents
- Visual representations such as charts and dashboards
- Business impact analysis (risk, cost, downtime)

Continuous Improvement:

Metrics are used to identify performance gaps. For example, a high Mean Time to Detect (MTTD) indicates delayed threat detection, requiring improvements in monitoring or alerting systems.

Practical Application

1. Threat Hunting Practice

Overview

This task focused on applying threat hunting techniques in a simulated SOC environment. The objective was to develop a hypothesis, analyze logs, validate Indicators of Compromise (IOCs), and map findings to attacker techniques.

Tools Used

- Elastic Security
- Velociraptor
- AlienVault OTX

Methodology

1. Hypothesis Development:

A hypothesis was created:

"Unauthorized privilege escalation may be occurring in domain accounts."

Log Analysis:

Queried security logs in Elastic Security focusing on **Event ID 4672**, which indicates special privilege assignment.

Observed Data:

Timestamp	User	Event ID	Notes
2025-08-18 15:00:00	testuser	4672	Unexpected admin role

This activity suggested suspicious privilege escalation.

2. Threat Intelligence Validation:

- Checked suspicious indicators using AlienVault OTX
- Identified potentially malicious IP patterns related to credential misuse

Endpoint Investigation:

- Used Velociraptor to query running processes

- Command used: `SELECT * FROM processes`
- Detected abnormal processes linked to suspicious activity

Findings

- A user account was assigned elevated privileges without proper authorization
- Suspicious behavior aligned with credential misuse patterns
- Indicators matched known threat intelligence data

MITRE ATT&CK Mapping

- Technique: **T1078 – Valid Accounts**
  This indicates attackers may be using legitimate credentials to gain unauthorized access and escalate privileges.

3. Hunting Report

During the threat hunting exercise, a hypothesis regarding unauthorized privilege escalation was tested using log analysis in Elastic Security. Event ID 4672 revealed suspicious privilege assignment to a user account. Threat intelligence validation through AlienVault OTX identified potential malicious indicators associated with credential misuse. Further investigation using Velociraptor confirmed the presence of abnormal processes. The activity was mapped to MITRE ATT&CK technique T1078 (Valid Accounts), indicating possible unauthorized use of legitimate credentials. This exercise demonstrated the effectiveness of proactive threat hunting in detecting hidden threats before they trigger alerts.

2. SOAR Playbook Development

Overview

This task focused on designing and testing an automated incident response workflow using SOAR tools. The objective was to reduce manual effort in handling phishing incidents by creating a playbook that performs validation, response, and case management automatically.

Tools Used

- Splunk Phantom
- TheHive
- Google Docs

Methodology

1. Playbook Design:

A playbook was created in Splunk Phantom to automate phishing incident response. The workflow included:

- Extract IP address from alert
- Check IP reputation using threat intelligence
- Block malicious IP using CrowdSec
- Generate incident case in TheHive

Playbook Workflow Steps:

1. Receive phishing alert from SIEM (Wazuh)
2. Enrich data by checking IP reputation
3. If malicious → trigger automated response
4. Block IP address
5. Create incident ticket for tracking

2. Playbook Testing:

A phishing alert was simulated in Wazuh to test automation. The playbook executed successfully and performed all defined actions.

Execution Results:

Playbook Step	Status	Notes
Check IP	Success	IP flagged as malicious
Block IP	Success	CrowdSec blocked 192.168.1.102

Findings

- Automated validation reduced manual investigation time
- Immediate IP blocking prevented further communication with malicious sources
- Incident tracking improved through automatic ticket creation

3. Playbook Summary

A SOAR playbook was developed using Splunk Phantom to automate phishing incident response. The playbook validated IP reputation, blocked malicious IPs using CrowdSec, and created incident tickets in TheHive. This automation reduced response time, minimized manual effort, and improved the overall efficiency of SOC operations.

3. Post-Incident Analysis

Overview

This task focused on analyzing a simulated phishing incident to identify its root cause, document lessons learned, and evaluate SOC performance using key metrics. The goal was to improve future detection and response strategies through structured analysis.

Tools Used

- Google Sheets
- Draw.io

Methodology

1. Root Cause Analysis (RCA):

The 5 Whys technique was applied to identify the underlying cause of the phishing incident.

RCA Findings:

Question	Answer
Why was the email opened?	User clicked malicious link
Why was link clicked?	Weak email filtering
Why was filtering weak?	Outdated security rules
Why were rules outdated?	Lack of regular updates
Why no updates?	No defined review process

2. Fishbone Diagram Analysis:

A Fishbone Diagram was created in Draw.io to categorize causes into:

- **People:** Lack of user awareness
- **Process:** No regular security review
- **Technology:** Weak email filtering system

3. Metrics Calculation:

- **Mean Time to Detect (MTTD):** 2 hours

- **Mean Time to Respond (MTTR):** 4 hours

These values indicate moderate detection capability but slower response handling.

Findings

- Primary cause was weak email filtering and lack of process updates
- User awareness alone is not the root problem
- Response time needs improvement to reduce impact

Metrics Summary

The incident analysis showed a Mean Time to Detect (MTTD) of 2 hours and a Mean Time to Respond (MTTR) of 4 hours. While detection was moderately efficient, response time was relatively slow. Improvements in automation and updated security controls can significantly enhance overall SOC performance.

4. Alert Triage with Automation

Overview

This task focused on analyzing security alerts and automating validation using threat intelligence. The objective was to prioritize alerts, reduce false positives, and improve response efficiency through integration of SIEM and incident response tools.

Tools Used

- Wazuh
- VirusTotal
- TheHive

Methodology

1. Alert Analysis (Triage):

A mock alert was generated and analyzed in Wazuh.

Alert Details:

Alert ID	Description	Source IP	Priority	Status
005	File Download	192.168.1.10	High	Open

The alert indicated a potentially suspicious file download from a local system.

2. Alert Investigation:

- Checked source IP behavior and activity patterns
- Identified abnormal file download activity
- Marked alert as high priority due to potential malware risk

3. Automated Threat Validation:

- Integrated TheHive with VirusTotal
- Automatically submitted file hash for analysis
- Retrieved reputation results without manual intervention

Findings

- The file hash was flagged as malicious by multiple antivirus engines
- Automation reduced analysis time significantly

- Alert validation became faster and more accurate

Automation Summary

The alert triage process was enhanced by integrating TheHive with VirusTotal for automated hash analysis. A suspicious file download alert was validated quickly, confirming malicious activity. Automation reduced manual effort, improved accuracy, and enabled faster prioritization and response to high-risk security incidents.

5. Evidence Analysis

Overview

This task focused on analyzing digital evidence collected from a system and maintaining proper chain-of-custody procedures. The objective was to identify suspicious activity from system data and ensure the integrity and admissibility of evidence.

Tools Used

- Velociraptor
- FTK Imager

Methodology

1. Evidence Collection:

System logs and network activity data were collected from a Windows virtual machine using Velociraptor. Additional disk-level evidence was acquired using FTK Imager to preserve data integrity.

2. Network Analysis:

Executed the following query in Velociraptor:

```
SELECT * FROM netstat
```

Observations:

- Identified active network connections
- Detected unusual outbound connections to unknown IP addresses
- Observed connections on uncommon ports, indicating potential malicious communication

3. Suspicious Findings

- Unknown external IP communication from internal system
- Persistent connections suggesting possible command-and-control (C2) activity
- Network behavior inconsistent with normal user activity

4. Chain-of-Custody Documentation

Item	Description	Collected By	Date	Hash Value
Network Log	Server-Z Log	SOC Analyst	2025-08-18	SHA256: A1B2C3D4E5F6

Findings

- Evidence indicated suspicious network communication
- Possible compromise involving unauthorized external connections
- Proper documentation ensured traceability and legal validity of evidence

6. Adversary Emulation Practice

Overview

This task focused on simulating real-world attacker behavior to evaluate the effectiveness of SOC detection mechanisms. The objective was to emulate a phishing attack using adversary emulation tools and verify whether security systems could detect and respond appropriately.

Tools Used

- MITRE Caldera
- Wazuh

Methodology

1. Attack Simulation:

A spear-phishing attack was simulated using MITRE Caldera based on MITRE ATT&CK technique **T1566 (Phishing)**. The simulation mimicked real-world attack behavior to test detection capabilities.

2. Detection Configuration:

Detection rules were configured in Wazuh to identify phishing-related indicators such as suspicious email activity and malicious links.

3. Detection Results:

Timestamp	TTP	Detection Status	Notes
2025-08-18 17:00:00	T1566	Detected	Phishing email blocked

Findings

- The simulated phishing attack was successfully detected
- Security controls were able to block the malicious activity
- Detection rules were effective but dependent on predefined patterns

Detection Gaps Identified

- Limited detection for highly obfuscated phishing payloads

- Dependence on signature-based detection rather than behavioral analysis
- Lack of advanced email anomaly detection

Emulation Report

An adversary emulation exercise was conducted using MITRE Caldera to simulate a spear-phishing attack (T1566). The attack was successfully detected by Wazuh, which generated alerts and enabled timely response. While the system effectively blocked the phishing attempt, the exercise revealed limitations in detecting advanced or obfuscated payloads. Detection relied primarily on predefined rules, indicating the need for behavioral analysis and improved email security mechanisms. This simulation highlighted the importance of continuous testing and tuning of detection systems to handle evolving attack techniques.

7. Security Metrics and Executive Reporting

Overview

This task focused on measuring SOC performance using key security metrics and presenting the findings in a format suitable for management. The objective was to analyze detection and response efficiency, identify gaps, and communicate insights clearly to non-technical stakeholders.

Tools Used

- Elastic Security
- Google Sheets
- Google Docs

Methodology

1. Metrics Calculation:

SOC performance metrics were calculated based on a simulated incident:

- **Mean Time to Detect (MTTD):** 2 hours
- **Mean Time to Respond (MTTR):** 4 hours
- **Dwell Time:** 6 hours (time between compromise and containment)

These metrics were used to evaluate the efficiency of detection and response processes.

2. Dashboard Creation:

A dashboard was created in Elastic Security to visualize:

- Detection and response timelines
- Alert trends
- False positive rates

This helped in identifying performance patterns and operational gaps.

3. Metrics Analysis

Findings:

- Detection time was moderate but can be improved with better monitoring
- Response time was relatively slow, indicating delays in containment
- Dwell time suggested attackers remained undetected for a significant duration

Metrics Analysis Summary

The analysis revealed a Mean Time to Detect (MTTD) of 2 hours and Mean Time to Respond (MTTR) of 4 hours, resulting in a dwell time of 6 hours. While detection was acceptable, response delays increased overall risk exposure, highlighting the need for faster incident handling and improved automation.

Executive Summary

The SOC performance analysis identified key areas for improvement in detection and response operations. The Mean Time to Detect (MTTD) was recorded at 2 hours, indicating moderate efficiency in identifying threats. However, the Mean Time to Respond (MTTR) was 4 hours, leading to a total dwell time of 6 hours, which increases the risk of system compromise. The analysis suggests that while monitoring systems are functioning effectively, response processes require optimization. It is recommended to implement automated response mechanisms, enhance threat intelligence integration, and regularly update detection rules to reduce response time. Additionally, continuous monitoring and periodic review of security controls will help improve overall SOC performance. Strengthening these areas will reduce risk exposure and enhance the organization's ability to respond to evolving cyber threats efficiently.

8. Capstone Project Comprehensive SOC Incident Response

Overview

This capstone project simulated a full-scale cyber attack and complete SOC workflow, including detection, triage, response, analysis, and reporting. The objective was to integrate all learned concepts into a real-world incident handling scenario.

Tools Used

- Metasploit
- Wazuh
- CrowdSec
- TheHive
- MITRE Caldera
- Elastic Security
- Google Docs

Methodology

1. Attack Simulation

A vulnerability in Metasploitable2 was exploited using Metasploit (Samba usermap script). This simulated unauthorized remote access.

2. Adversary Emulation

A related attack technique T1210 (Exploitation of Remote Services) was simulated using MITRE Caldera.

Detection Log:

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 16:00:00	192.168.1.10	Samba exploit	T1210

3. Detection and Triage

- Wazuh generated alerts for suspicious activity
- Alerts were triaged in TheHive
- Incident was classified as **High Severity** due to exploitation attempt

4. Response and Containment

- Infected VM was isolated from the network

- Malicious IP (192.168.1.102) was blocked using CrowdSec
- Containment was verified using a ping test (no response from attacker IP)

5. SOAR Automation

- A case was created in TheHive
- Automated playbook executed to block IP and log incident
- Reduced manual response time

6. Post-Incident Analysis

- Root Cause Analysis performed using 5 Whys technique
- Identified cause: vulnerable service with weak security configuration
- Fishbone Diagram created to analyze contributing factors

7. Metrics Reporting

- **MTTD:** 2 hours
- **MTTR:** 4 hours
- **Dwell Time:** 6 hours

Dashboard created in Elastic Security to visualize SOC performance.

Final Incident Report

A simulated cyber attack was conducted using Metasploit to exploit a Samba vulnerability in a Metasploitable2 environment. The attack represented a real-world exploitation scenario mapped to MITRE ATT&CK technique T1210 (Exploitation of Remote Services). Wazuh successfully detected the attack and generated alerts, which were triaged in TheHive and classified as high severity.

Immediate response actions included isolating the affected system and blocking the attacker's IP address using CrowdSec. Automation was implemented through a SOAR playbook, reducing response time and ensuring consistent incident handling. Post-incident analysis identified the root cause as a vulnerable service and lack of proper patch management.

Performance metrics showed a Mean Time to Detect (MTTD) of 2 hours and Mean Time to Respond (MTTR) of 4 hours, resulting in a dwell time of 6 hours. While detection mechanisms were effective, response time indicated scope for improvement through enhanced automation and faster decision-making.

It is recommended to implement regular vulnerability assessments, improve patch management processes, and strengthen detection rules. Continuous monitoring and periodic security testing should be conducted to prevent similar incidents. This project demonstrated the effectiveness of an integrated SOC approach in detecting, responding to, and analyzing cyber threats.

Stakeholder Briefing

A simulated cyber attack was conducted to evaluate the organization's security response capabilities. The attack exploited a system vulnerability and was detected within 2 hours. The response team contained the threat within 4 hours by isolating the affected system and blocking the attacker's access.

The analysis revealed that the primary issue was an unpatched vulnerability, highlighting the need for better system maintenance and monitoring. While detection systems performed effectively, response time can be improved through increased automation and faster decision-making processes.

To strengthen security, it is recommended to implement regular system updates, enhance monitoring tools, and conduct periodic security testing. Overall, the organization demonstrated a strong ability to detect and respond to threats, but improvements are required to reduce response time and minimize risk exposure.